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(Autonomous)

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EXPT NO: 3	DATE: _______________

A PYTHON PROGRAM TO IMPLEMENT THE ASSIGNMENT PROBLEM

AIM:

To solve the Assignment Problem using the Hungarian Algorithm (linear_sum_assignment) in Python to find the optimal assignment of tasks to workers minimizing or maximizing total cost/profit.

PROCEDURE:

1. Install numpy and scipy.
2. Define the cost matrix where cost[i][j] = cost of assigning worker i to task j.
3. For maximization, subtract all values from the maximum element to convert to cost.
4. Apply scipy.optimize.linear_sum_assignment to find optimal row and column indices.
5. Compute and display total cost/profit.

PROBLEM 1: 4 Workers x 4 Tasks – Minimization

CODE

```
import numpy as np
from scipy.optimize import linear_sum_assignment
cost = np.array([[9, 2, 7, 8],[6, 4, 3, 7],[5, 8, 1, 8],[7, 6, 9, 4]])
row_ind, col_ind = linear_sum_assignment(cost)
print("Optimal Assignment:")
for r, c in zip(row_ind, col_ind):
print(f" Worker {r+1} -> Task {c+1} (cost={cost[r,c]})")
print("Minimum Total Cost:", cost[row_ind, col_ind].sum())
```

OUTPUT

```
Optimal Assignment:
Worker 1 -> Task 2 (cost=2)
Worker 2 -> Task 3 (cost=3)
Worker 3 -> Task 1 (cost=5)
Worker 4 -> Task 4 (cost=4)
Minimum Total Cost: 14
```

PROBLEM 2: 3 Workers x 3 Tasks – Maximization (Profit Matrix)

CODE

```
profit = np.array([[40, 60, 20],[50, 30, 70],[20, 70, 50]])
cost = profit.max() - profit
row_ind, col_ind = linear_sum_assignment(cost)
print("Optimal Assignment:")
for r, c in zip(row_ind, col_ind):
print(f" Worker {r+1} -> Job {c+1} (profit={profit[r,c]})")
```

```
print("Maximum Total Profit:", profit[row_ind, col_ind].sum())
```

OUTPUT

```
Optimal Assignment:
Worker 1 -> Job 2 (profit=60)
Worker 2 -> Job 3 (profit=70)
Worker 3 -> Job 1 (profit=20)
Maximum Total Profit: 150
```

PROBLEM 3: 5 Workers x 4 Tasks – Unbalanced Minimization

CODE

```
cost = np.array([[3,7,5,4],[2,3,1,6],[4,1,6,5],[7,8,9,2],[6,5,4,7]])
padded = np.pad(cost, ((0,0),(0,1)), constant_values=0)
row_ind, col_ind = linear_sum_assignment(padded)
print("Optimal Assignment (unbalanced):")
for r, c in zip(row_ind, col_ind):
if c < cost.shape[1]:
print(f" Worker {r+1} -> Task {c+1} (cost={cost[r,c]})")
else:
print(f" Worker {r+1} -> Unassigned")
valid = [(r,c) for r,c in zip(row_ind,col_ind) if c < cost.shape[1]]
total = sum(cost[r,c] for r,c in valid)
print("Minimum Total Cost:", total)
```

OUTPUT

```
Optimal Assignment (unbalanced):
Worker 1 -> Task 4 (cost=4)
Worker 2 -> Task 3 (cost=1)
Worker 3 -> Task 2 (cost=1)
Worker 4 -> Unassigned
Worker 5 -> Task 1 (cost=6)
Minimum Total Cost: 12
```

RESULT:

Thus, the Assignment Problem using the Hungarian Algorithm in Python has been implemented and executed successfully.